Candidate Seat No. ....

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech Theory Examination (Mar-Apr 2018)

MA202 Engineering Mathematics III

Date: 26.03.2018, Monday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper contains two sections. Each section has three questions.
2. Section I and II must be attempted in separate answer sheets.
3. Each section carries 35 marks.
4. Figures to the right indicate marks.
5. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Multiple Choice Questions **[07]**

(1) If the given function $f(x)$ is an odd function, then it's Fourier series expansion contains ____________________________.

(i) Only cosine terms (ii) Only sine terms
(iii) Both sine and cosine terms (iv) None of these

(2) The inverse Laplace Transform of $\dfrac{k}{s^2+k^2}$ = __________________

(i) $\sin kt$ (ii) $\cos kt$
(iii) $H(t-a)\sin kt$ (iv) None of these

(3) $L\{f'(t)\} =$ _________________

(i) $\overline{f}(s)$ (ii) $s\overline{f}(s)$
(iii) $s\overline{f}(s) - f(0)$ (iv) Not defined

(4) The function $f(x) = x^2; 0 < x < 2$ is ________________ function.

(i) even (ii) odd
(iii) periodic (iv) neither even nor odd

(5) The period of $\cos pt$ is ________________

(i) 2π (ii) $\dfrac{\pi}{p}$
(iii) $\dfrac{2\pi}{p}$ (iv) None of these

(6) $L^{-1}\{\overline{f}(s)\overline{g}(s)\} =$ _________________.

(i) $f(t)g(t)$ (ii) $f(t)*g(t)$
(iii) $f(t)+g(t)$ (iv) $\overline{f}(s)*\overline{g}(s)$

(7) Laplace transform is defined only for ______________ functions of t .

(i) positive (ii) negative
(iii) positive and negative (v) none of these

Q - 2 Attempt any ***four*** out of ***six*** **[16]**

1) Solve $\frac{d^2y}{dt^2}+y=\sin 2t, y(0)=0, y'(0)=0$ using Laplace transform.

2) Find the half range cosine series of $f(x)$,

where $f(x)=x \qquad 0<x<\frac{\pi}{2}$

$\qquad =\pi-x \qquad \frac{\pi}{2}<x<\pi$

3) Solve $x^4+2x^3-x^2-2x-3=0$ by Ferrari's method.

4) State Convolution theorem and evaluate $L^{-1}\left\{\frac{a}{s^2(s^2+a^2)}\right\}$.

5) Find the Fourier series of $f(x)=1-x^2$ in the interval (-1,1).

6) Find $(i)\ L\{(t+1)^2e^t\} \quad (ii)\ L\left\{\frac{\sin 2t}{t}\right\}$

Q – 3 (a) Attempt any ***two*** out of ***three*** **[06]**

(1) Find the Fourier series of $f(x)=x$ in the interval $(0,2\pi)$.

(2) Find Laplace transform of $f(t)=e^t$, $0<t<2\pi$ if $f(t)=f(t+2\pi)$.

(3) Solve $x^3+6x^2+11x+6=0$ by Cardon's method.

Q – 3 (b) Attempt any ***two*** out of ***three*** **.** **[06]**

(1) Obtain the Fourier cosine series for function $f(x)=e^x$ in the range $(0,l)$.

(2) Find the inverse Laplace transform of $\frac{s+2}{s(s+1)(s+3)}$.

(3) Solve $x^3-27x+54=0$ by Cardon's method.

SECTION – II

Q – 4 Multiple Choice Questions **[07]**

1) The expression curl(grad *f*),where *f* is a scalar function, is

(a) equal to $\nabla^2 f$ (b) equal to div(grad *f*) (c) a scalar of zero magnitude

(d) a vector of zero magnitude

2) The Gauss divergence theorem relates certain surface integrals to …………..

1) Line integral (b) Volume integral (c) Surface integral (d) None of above

3) If $\vec{r}=(x+3y)\hat{i}+(2x-3y)\hat{j}+(x+az)\hat{k}$ is solenoidal vector then the value of a is

(a) 0 (b) 1 (c) 2 (d) 3

4) The line integral $\int_A^B \vec{F}d\vec{r}$ be independent of the path if …………….

(a) $div\vec{F}=0$ (b) $curl\vec{F}=0$ (c) $div\vec{F}\neq 0$ (d) $curl\vec{F}\neq 0$

5) Let f(x,y,z)=c represent the equation of the surface. Then unit normal vector to this surface is

(a) $\frac{grad\ f}{|grad\ f|}$ (b) $grad\ f$ (c) $div(grad\ f)$ (d) $curl(grad\ f)$

6) If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ then $curl\ \vec{r}$ equal to

(a) 3 (b) r (c) 0 (d) -r

7) A vector $\vec{F}$ is solenoidal if …………

(a) $div\vec{F}=0$ (b) $curl\vec{F}=0$ (c) $div\vec{F}\neq 0$ (d) $curl\vec{F}\neq 0$

Q -5 Attempt any ***four*** out of ***six*** **[16]**

1) Show that $\vec{F}=\left(y^2+2xz^2\right)\hat{i}+\left(2xy-z\right)j+\left(2x^2z-y+2z\right)\hat{k}$ is irrotational and hence find its scalar potential.

2) A particle moves along the curve $x=2t^2, y=t^2-4t, z=3t-5$,where t is the time. Find the components of its velocity and acceleration at time t=1 in the direction $\hat{i}+2j+2k$.

3) Evaluate $\int_C y^2dx+xdy$,where C is the arc of parabola $x=4-y^2$ between (-5 , -3) and (0 , 2).

4) Use Green's theorem, to evaluate $\oint_C xydx+x^2y^3dy$, where C is the triangle with vertices (0 , 0) , (1 , 0) and (1 , 2).

5) An e.m.f. Esinpt is applied at t=0 to a circuit containing a condenser of capacity C and inductance L in series. The current *i* satisfies the equation $L\frac{di}{dt}+\frac{1}{C}\int idt=E\sin pt$. If $p^2=\frac{1}{LC}$ and initially the charge q and the current *i* are zero,show that the current at time t is $\frac{Et}{2L}\sin pt, i=\frac{dq}{dt}$.

6) A homogeneous rod of conducting material of length 100 cm. has its ends kept at zero temperature and the temperature initially is

$$u(x,0)=x, 0\le x\le 50$$
$$=100-x, 50\le x\le 100$$

where equation of heat conduction is $\frac{\partial u}{\partial t}=c^2\frac{\partial^2 u}{\partial x^2}$

find the temperature u(x,t) at any time.

Q –6 (a) Attempt any ***two*** out of ***three*** **[06]**

1) Find the directional derivative of $f(x,y,z) = 2x^2 + 3y^2 + z^2$ at the point P(2 ,1 ,3) in the direction of the vector $\overset{\frown}{a} = \hat{i} - 2\overset{\frown}{k}$.

2) Using Divergence theorem, evaluate the surface integral $\iint_S (yzdydz + zxdzdx + xydydx)$ where $S : x^2 + y^2 + z^2 = 4$.

3) Show that the frequency of free vibrations in a closed electrical circuit with inductance L and capacity C in series is $\dfrac{30}{\pi\sqrt{LC}}$ cycles/minute.

Q –6 (b) Attempt any ***one*** out of ***two*** **[06]**

1) Find the directional derivative of $\nabla(\nabla f)$ at the point (1 ,-2 ,1) in the direction of the normal to the surface $xy^2z = 3x + z^2$ where $f = 2x^3y^2z^4$

2) Verify Stoke's theorem for $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\, j$ in the rectangular region bounded by $x = 0, x = a, y = 0, y = b$